



Grounds for comparison: linking Arabica cup scores to soil fertility and plant nutrition in rust-resistant cultivars and traditional American coffees

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Received: 21 March 2026 / Accepted: 1 September 2026
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Abstract The cup quality of rust-resistant Arabica (*Coffea arabica* L.) × Robusta (*C. canephora* Pierre ex Froehner) cultivars is thought to be inferior to traditional Arabicas, but few studies have examined the extent to which coffee cup scores are affected by local variation in soil fertility and plant nutrition. We evaluated coffee cup quality using the Specialty Coffee Association's 100-point Q-grading system and characterized the timing and formulation of soil and foliar nutrient additions across 23 fields in south-central Guatemala managed by smallholder farmers, then compared relationships between soil extractable and total leaf and green bean nutrient concentrations and

coffee cup scores for traditional versus rust-resistant Arabica cultivars. All cultivars had similar coffee cup qualities and obtained final cup scores suitable for special coffee market access (> 80.0 points). The timing of nutrient additions closely aligned with local weather conditions, but formulations of foliar nutrient additions differed markedly between cultivars. In > 80% of fields, soil inorganic N, K, Mg and B and leaf S, Zn, Cu, and B concentrations were below minimum thresholds generally ideal for crop production. Despite potential nutrient deficiencies, most cup qualities were positively correlated with leaf Zn for traditional cultivars ($R^2_{\text{adj}} = 0.23\text{--}0.82$) and soil K for rust-resistant cultivars ($R^2_{\text{adj}} = 0.34\text{--}0.78$). Our study demonstrates rust-resistant Arabicas grown in this region of Guatemala can have similarly appealing coffee cup flavor profiles and organoleptic qualities as Bourbon and Typica lineages. However, a larger network of sites is needed to establish reliable soil and/or foliar indicators of coffee cup quality across variable management practices and soil conditions.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10705-026-10525-w>.

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Keywords Coffee · Coffee arabica · Cup quality · Andisol

Introduction

At 2.4 billion cups consumed daily (ICO 2024a) and \$36.3 USD billion traded annually (Workman 2023), coffee is one of the most popular beverages in the

world and the most globally commodity crops of the tropics (Mussatto et al. 2011; Pancsira 2022). Arabica (*Coffea arabica* L.) represented 70% of total global commercial coffee production in the 2023/24 crop year (ICO 2024b). Approximately 80% of global Arabica coffee exports are grown by smallholder farmers in Latin America (Kishaija et al. 2025), generally geographically isolated, remote regions in steeply sloped mountainous landscapes (≈ 1000 – 2000 m asl) (Harvey et al. 2021). Due to the lack of protective mechanisms for tropical commodity markets (e.g., farm insurance) (Barnett and Adger 2007; Tucker et al. 2010) and large-scale roasters and retailers weakening coffee producing countries' ability to self-regulate export prices (Grabs and Ponte 2019), stagnant or negative profit margins continue to compromise the livelihoods and economic security of resource-constrained coffee farmers in Latin America (Daviron and Ponte 2005; Eakin et al. 2006; Barreto Pexixoto et al. 2023).

Unlike low-priced, poor quality commodity blends (Pendergrast 2010), the superior cup quality of single-origin coffee sold in specialty markets appeals to consumers willing to pay high prices for coffee with unique flavor profiles and sensory characteristics (Grabs and Ponte 2019). The volcanic highlands of Central America are widely known for producing some of the world's best tasting coffees (McCook and Montero-Mora 2024). Here, coffee is predominately grown by smallholder farmers in shaded agroforestry systems at elevations (≈ 700 – 2200 m asl) and air temperatures (≈ 18 – 21 °C) optimal for Arabica productivity (van der Vossen et al. 2015), proper cherry development maturation (Perfecto and Vandermeer 2002), and favorable cup qualities (Bosselmann et al. 2009; DaMatta et al. 2012; Sánchez-Hernández et al. 2018). Coffee farmers tend to prioritize management knowledge and cup quality over pest and disease management costs in cultivar selection (Turreira-García 2022). As a result, many farmers prefer traditional cultivars known for producing high quality coffee beans with high cup quality (e.g., Caturra) despite greater susceptibility to coffee leaf rust (*Hemileia vastatrix*) and other pathogens (e.g., *Cercospora coffeicola*) (McCook and Montero-Mora 2024). In contrast, rust-resistant cultivars derived from Arabica $\times$ *C. canephora* progenies are generally considered to produce inferior cup quality (Tay 2025), in particular bitter and astringent flavors in some

of the early Catimors (McCook 2019; McCook and Montero-Mora 2024). However, recently developed, first-generation (F_1) hybrids of rust-resistant cultivars have been found to obtain higher cup scores than traditional cultivars across a range environmental conditions (Marie et al. 2020).

The extent to which soil fertility impacts cultivar-specific coffee cup quality remains largely unclear and understudied across the global Coffee Belt (Geeraert et al. 2019; Tassew et al. 2021; Mintesnot et al. 2015) and within Latin America specifically (Arenas-Clavijo and Armbrrecht 2018; Rekik et al. 2019; Márquez et al. 2020). The general lack of field-tested, evidence-base fertilizer recommendations for tropical perennial crops translates challenges economically optimized fertilizer use, in contrast to many annual crops (e.g., Wang et al. 2015; Eash et al. 2019), further compromising the economic stability of resource-limited coffee farmers. Soil properties can impose constraints on soil nutrient availability (e.g., low cation exchange capacity (CEC) of coarse textured soils) that may indirectly influence coffee cup quality (Mitchell 1988; Feria-Morales 2002; Läderach 2007). However, soil tests used to prescribe soil fertility at 0–15 cm depth may offer less diagnostic insight into deep-rooted perennial crops like coffee compared to leaf nutrient concentrations used to gauge plant nutrition (Stephens 1968; Wang et al. 2015; Kasmerchak et al. 2025). Establishing such relationships between cup scores and measures of soil fertility and plant nutrition would help expand smallholder access to specialty coffee markets and tailor management practices to better optimize cup quality.

We evaluated potential relationships between coffee cup scores and soil fertility and/or plant nutrition in 23 coffee fields under realistic conditions of resource-limited, low input coffee production in the south-central Guatemalan highlands across highly similar soil types and growing conditions (Brown 2019; Allen Asensio et al. 2024). Smallholder farmers in this region currently represent 97% of coffee producers in Guatemala and generate 44% of country's coffee harvest (Tay 2024). Our specific objectives were to (i) assess coffee cup quality for Arabica cultivars traditionally grown in Central America and rust-resistant Arabica $\times$ *C. canephora* cultivars, (ii) characterize nutrient management practices and soil fertility (e.g., pH, extractable nutrients) and plant nutrition (i.e., leaf nutrient concentrations), and (iii)

determine the extent to which measures of soil fertility and plant nutrition potentially explain cultivar-specific variation in coffee cup qualities and final cup scores. We hypothesized (1) traditional Arabica cultivars would obtain higher cup scores than rust-resistant cultivars, regardless of (2) variable management practices and plant nutrition status across similar soil nutrient limitations. We also hypothesized (3) cup scores would exhibit similarly weak correlations with soil fertility measures compared to strong cultivar-specific relationships with plant nutrition measures.

Methods

Study area and design

We tested hypotheses on cup quality $\times$ soil fertility $\times$ plant nutrition relationships across 23 coffee fields managed by 23 farmers in the highlands of south-central Guatemala in the township of San Pedro Yepocapa within the Department of Chimaltenango, Guatemala, Hermogenes Montellano (14°29'57.88" N; 90°59'31.47" W). Coffee production in this region is representative of the relatively low-input coffee production systems historically found in the steeply-sloped volcanic highlands of Central America (Harvey et al. 2018, 2021). The regional climate is classified as sub-tropical highland (Cwb Koppen classification), with mean annual temperature ranging 22–28 °C and 2500–3000 mm of rainfall, largely concentrated from May to October. Soils in this region formed in pyroclastic material from the active Volcán de Fuego stratovolcano located $\approx$ 12 km to the southwest of our sampling region. Analysis of ash samples collected during a period of high volcanic activity in 2018 (Albino et al. 2020; Charbonnier et al. 2023) suggests ash deposition potentially provides appreciable Ca, Cu, Mn, and Zn inputs (Table S1). However, high sand content (> 70%), steep slopes (30°–68°), and relatively low cation exchange capacity (CEC) (median 6.0 meq 100 g⁻¹) (Allen Asensio et al. 2024), entails high infiltration rates and entails substantial nutrient leaching under the relatively high rainfall in this region (Allen Asensio et al. 2026).

We selected fields planted with Arabica cultivars representative of smallholder coffee production in Central America (Tay 2025): (1) traditional cultivars commonly grown in Central America ascended

from Bourbon and Typica lineages, including Bourbon, Caturra, Pacas, and Cautai, (n=9 fields), and (2) rust-resistant cultivars derived from Arabica $\times$ *C. canephora* progenies, including Sarchimors, Catimors, Icatú, and Centroamericano (n=14 fields) (Table 1). With farmer consent, we selected fields from a comprehensive 2018–2019 baseline soil fertility assessment that revealed prevalent soil nitrogen (N) and potassium (K) deficiencies throughout region (Allen Asensio et al. 2024). Fields needed to have at least 24 mature coffee plants of uniform cultivar that were actively bearing fruit for at least 3 y post-establishment. Coffee was planted at densities ranging from 2000 to 3333 trees ha⁻¹ (mean = 2975 $\pm$ 91 trees ha⁻¹) under varying degrees of shade from native and introduced tree species (e.g., *Grevillea robusta*) and common agroforestry species used for fruit crops (e.g., *Persea americana*), N-fixation (e.g., *Inga edulis*), and timber (e.g., *Cedrus libani*) planted at 138–625 trees ha⁻¹ (4:1 to 24:1 coffee-to-shade ratio) (Table S2). We expected local variation in shade tree density in the high-elevation fields in our study ($\approx$ 800–1600 m asl) to minimally impact coffee cup quality compared to low-elevation systems where cherry maturation is highly sensitive to relative shade cover (< 700 m asl) (Guyot et al. 1996; Salazar et al. 2000; Muschler 2001).

Sample collection and processing

We harvested coffee cherries, sampled coffee leaves and soils, and conducted on-farm interviews during the same visit to each field from December 2019 to January 2020, prior to fertilizer application in April. This timing was chosen to coincide with peak harvest when cherries are most likely to have optimal quality attributes (i.e., ripeness, exterior conditions indicative of minimal defects), thus maximizing cup quality potential. We harvested approximately double the mass of ripe coffee cherries needed for processing ($\approx$ 2.2 kg) and used a hand-crank depulper to remove exocarps from cherries of optimal ripeness and no physical signs of defects within 4 h post-harvest. To mimic the wet processing (i.e., washed) method most used by farmers in this region (Figuerola Campos et al. 2020), we fermented depulped cherries in unsealed 1-gallon sealable plastic bags for 48 h without any additional water (excluding low density “floater” cherries), then removed all visible mucilage

Table 1 Characteristics of Arabica coffee (*Coffea arabica* L.) fields and cultivars in south-central Guatemala that were evaluated for cup quality, soil fertility, and plant nutrition (n=23 fields). Fields were dominated by cultivars ascended from (i)Bourbon and Typica lineages commonly grown in Central America (Bourbon, Caturra, Pacas, Catuái), or (ii) rust-resistant cultivars derived from Arabica × *C. canephora* progenies (Sarchimors, Catimors, Icatú, Centroamericano)

ID	Elevation (m asl)	Field area (ha)	Planting density (trees ha ⁻¹)	Cultivar sampled	Genetic parentage ^a
1	1280	4.1	3333	Catuái	Caturra × Mundo Novo
2	1230	3.9	2941	Sarchimors	Villa Sarchi × Híbrido de Timor
3	1210	3.9	2000	Centroamericano	Sarchimor T5296 × Rume Sudan landrace
4	1190	2.8	2778	Caturra	Bourbon
5	1200	2.8	3333	Centroamericano	Sarchimor T5296 × Rume Sudan landrace
6	1335	4.1	3333	Centroamericano	Sarchimor T5296 × Rume Sudan landrace
7	1380	4.1	3333	Caturra	Bourbon
8	1390	3.1	2500	Centroamericano	Sarchimor T5296 × Rume Sudan landrace
9	1315	4.1	3125	Icatú	Robusta × Bourbon Vermelho × Mundo Novo BC
10	1315	4.1	3333	Centroamericano	Sarchimor T5296 × Rume Sudan landrace
11	1315	4.1	3333	Caturra	Bourbon
12	1315	4.1	2000	Bourbon	Typica
13	1600	4.1	2778	Pacas	Bourbon
14	1325	4.1	3333	Sarchimors	Villa Sarchi × Híbrido de Timor
15	1290	4.1	3333	Pacas	Bourbon
16	1305	3.9	2778	Icatú	Robusta × Bourbon Vermelho × Mundo Novo BC
17	750	1.4	3333	Catimors	Caturra × Híbrido de Timor
18	1260	4.1	3086	Catimors	Caturra × Híbrido de Timor
19	1285	2.4	2500	Centroamericano	Sarchimor T5296 × Rume Sudan landrace
20	1330	4.1	3333	Pacas	Bourbon
21	1190	1.8	2500	Catimors	Caturra × Híbrido de Timor
22	1290	3.9	3333	Centroamericano	Sarchimor T5296 × Rume Sudan landrace
23	1315	0.9	2778	Pacas	Bourbon

^aAccessed from World Coffee Research (WRC) catalog (WRC 2024)

particles using local tap water. Washed cherries were then sun-dried screens or tarps until reaching 10–12% moisture content (7–10 d), then dehulled using a table-top electric dehulling machine to remove the endocarp layer. Following Specialty Coffee Association (SCA) protocols (SCA 2020a), we manually inspected the remaining green (i.e., unroasted) coffee beans for physical defects (e.g., insect damage, breakage, discoloration, mechanical damage) and removed small beans using a coffee screen (1.98 mm). A sub-sample of green beans (5 g) was ground (<0.1 mm) for nutrient analysis using a mechanical stainless-steel mill and prepared for shipment.

During cherry harvest, we collected opposing pairs of mature leaves from the middle section of four branches at the mid-line of the tree (roughly chest height; 8 total leaves per plant) from three plants

per field at each cardinal direction. Leaf samples were then composited and stored in paper bags in open air to dry in at ambient conditions then ground (<0.1 mm) for nutrient analysis using a mechanical stainless-steel mill and prepared for shipment. Below the same coffee trees selected for cherry harvesting, we sampled soils (0–15 cm depth) using a hand auger at eight randomized points within each field, which were then composited before air-drying and sieving (<2 mm). During the same visit, we conducted on-site interviews to characterize nutrient management practices implemented by local farmers over the growing season. Farmers were asked to describe the timing, frequency, and formulation of foliar nutrient additions and soil fertilizer inputs. Interviews took place within each farmer's field and were carried out by two personnel who were familiar with the farming

community. Interview responses detailing farmer perceptions of management objectives were not included in the present study, as the focus was on soil fertility and plant nutrition relationships with cup quality.

Data collection

We used SCA cup grading protocols to ensure our assessment of coffee cup quality was objective and reflected current standards used to differentiate specialty market prices (SCA 2020b). Briefly, a roasting expert used a gas-flame-heated sample roaster to whole green coffee beans processed from harvested cherries. Roasted coffee beans were then ground and steeped in ceramic coffee cupping bowls with non-distilled water (125–175 ppm total dissolved solids) and heated to 92.2–94.4 °C. Three Coffee Quality Institute (CQI) certified Arabica coffee Q-graders characterized flavor profiles and tasting notes, then assigned scores to aroma, flavor, aftertaste, acidity, body, balance, uniformity, clean cup, sweetness, and overall quality attributes to three brewed cups (modified from the five-cup SCA protocol) (Table S3). Scores assigned to these then cup qualities were then summed to obtain a final cup score on the 100-point SCA Q-grading scale. We differentiated the coffee cup quality using final cup score ranges reported in the specialty coffee transaction guide (Roberts and Treweek 2024): (i) 81.0–81.99 points, (ii) 82.0–83.99 points, (iii) 84.0–85.99 points, (iv) 86.0–86.9 points, and (v) > 87.0 points. Although these intervals correspond with free on board (i.e., pre-export) specialty coffee price estimates, a robust economic analysis describing the earning potential of coffee produced in this region was beyond the scope of our study, thus free on-board prices are not discussed directly.

Coffee leaf and green (i.e., unroasted) bean subsamples were digested in nitric acid (90 °C) and analyzed for total concentrations of phosphorus (P), K, calcium (Ca), magnesium (Mg), boron (B), zinc (Zn), copper (Cu), manganese (Mn), iron (Fe), and aluminum (Al) using inductively coupled plasma optical emission spectroscopy (ICP-OES) (Zarcinas et al. 1987) and total N by dry combustion. The agronomic and cup quality focus of our research meant characterizing green bean primary metabolite profiles and aromatic compound concentrations related to Central American coffee cup quality (e.g., Marie et al. 2020) was out of scope. Mehlich-3 extractable soil nutrient

concentrations were quantified on air-dried and sieved soils for P, K, Ca, Mg, B, Zn, Cu, Fe, and Al using ICP-OES (Ziadi and Sen Tran 2007). Following extraction by 0.5 M K₂SO₄ for 30 min using a 1:5 soil mass:extract volume ratio (Nelson 1983), inorganic N concentrations were quantified for ammonium-N (NH₄⁺-N) by salicylate using nitroprusside as a catalyst (Kempers and Zweers 1986) and NO₃⁻-N by vanadium (III) reduction (Hendrix and Braman 1995; Doane and Horwath 2003). We estimated soil cation exchange capacity (CEC) as the sum soil Ca, Mg, K, and sodium concentrations (Mattila and Rajala 2022), and measured soil pH using a pH probe on 1:1 soil mass:water slurry (USDA NRCS 2004). Total soil organic carbon (SOC) and total soil N (TSN) were quantified by dry combustion using a total elemental analyzer following HCl fumigation to remove potential carbonates (Harris et al. 2001).

Statistical analyses

To test our first and second hypotheses, we compared the distributions of coffee cup qualities, final cup scores, and soil fertility and plant nutrition measures for traditional versus rust-resistant cultivars using a Wilcoxon rank-sum test (i.e., non-parametric t-test). In R-Studio we used the *pairwise.wilcox.test()* function followed by the Benjamini–Hochberg false-discovery rate (FDR) *p*-value adjustment method in the {stats} package within base R (v4.2.1; R Core Team 2022). On-farm interview responses for the timing, frequency, and formulation of foliar nutrition additions and soil fertilizer inputs were analyzed qualitatively to better characterize between-cultivar differences in nutrient management practices and inform potential variation in soil extractable and total leaf/green bean nutrient concentrations. We assessed coffee plant nutrition using minimum leaf nutrient concentration thresholds recommended for optimal Arabica coffee production in Central America (Llatch-Hill 2017); no minimum concentrations thresholds are currently reported for green bean nutrients. Due to the conspicuous absence of peer-reviewed, evidence-based soil fertility recommendations for coffee (van Asten et al. 2010; Bhattarai et al. 2017; Wang et al. 2015) or woody perennials in Latin America in general (Chavez et al. 2021), we used soil fertility thresholds developed for annual crops in U.S.

agroecosystems a general basis for estimating relative nutrient deficiency for optimal crop production.

To test our third hypothesis, we first calculated Pearson's correlation coefficients (PCCs) for cup qualities and final cup scores across individual measures of soil fertility ($n=19$) and plant nutrition ($n=12$ for leaf and green beans) in R-Studio using the *cor()* function in {Hmisc} package (v.4.7.2; Harrell 2025). Univariate relationships were then visualized as correlation matrices generated using *corrplot()* function in the {corrplot} package (v.0.92; Wei and Simko 2024). To determine which variable(s) the most important predictors of coffee cup qualities and final cup scores for traditional versus rust-resistant cultivars, we constructed least absolute shrinkage and selection operator (LASSO) regression models using the *cv.glmnet()* command in the {glmnet} package (v4.1.8; Friedman et al. 2010). The tuning parameter (λ) the LASSO regression penalization function shrinks β coefficients equally (based on their absolute sum) as the number of predictor variables increase in size (McNeish 2015), and predictor variables with β coefficients shrunk to zero discarded from the final model (Tibshirani 1996). Thus, LASSO regression more effectively minimizes the bias-variance trade-off and over-fitting commonly observed least squares variable selection methods (Tibshirani 1996; James et al. 2017). We used k -fold cross-validation ($k=10$) with 1000 iterations of the tuning parameter (λ) with the lowest mean-square error to ensure standardized β coefficients were not overly penalized (Zou and Hastie 2005; Zou et al. 2007), then compared the adjusted correlation coefficients ($R^2_{\text{adj.}}$) and root-mean square errors (RMSE) of final LASSO models to best-fit univariate regression models to avoid over-fitting. All data visualizations were generated as vector files using the {tidyverse} collection (v2.0.0; Wickham et al. 2019), then exported and edited in Adobe Illustrator (v26.3.1; Adobe Inc. 2019).

Results

Coffee cup qualities and final scores

Traditional and rust-resistant cultivars obtained final cup scores suitable for specialty coffee market access (i.e., >80 points), though with appreciable within-cultivar variation among cup qualities

(Fig. 1). Excluding perfect scores for uniformity, sweetness, and clean cup across cultivars, cup qualities were scored lower on average for Pacas and Icatú (7.5 points) compared to Caturra and Centroamericano (mean = 7.8 points). Although cup quality scores were similar ($P=0.33$ – 0.75), traditional cultivars exhibited markedly greater variation across cup qualities on average (coefficient of variation [CV] = 10%) compared to rust-resistant cultivars (CV = 1%). Rust-resistant cultivars also obtained similar final cup scores as traditional cultivars known for high cup quality ($P=0.49$). However, a larger proportion of rust-resistant cultivars (43%) obtained final cup scores capable of fetching higher free on-board specialty coffee prices (84–85.99 points) compared to traditional cultivars (22%).

On-farm management and soil fertility and plant nutrition

Farmers responses to on-farm interviews suggest the timing and formulation of foliar nutrient additions varied appreciably between cultivars over the growing season compared to soil fertilizer inputs (Fig. 2). Specifically, farmers reported applying foliar nutrients to traditional cultivars throughout the growing season and to rust-resistant cultivars predominately from May to July (Fig. 2A). All farmers added foliar nutrients predominately as N or K or NPK mixtures (≈ 10 – 35% of farmers), but fewer farmers foliar B isolates or BZn or BCa mixtures to traditional ($\approx 10\%$ max) versus rust-resistant cultivars ($\approx 35\%$ max in May). Fewer farmers also reported applying foliar nutrients of unknown composition (e.g., locally fermented plant debris, proprietary blends) to traditional cultivars ($\approx 22\%$ max in August) compared to rust-resistant cultivars ($\approx 35\%$ max in mid-June). Nearly all farmers (95%) applied soil fertilizer as NP or NPK mixtures at the onset of the rainy season (May to June) (Fig. 2B). Farmers applied a second round of soil fertilizer during the peak growing season when growing traditional cultivars (September to October), and more flexibly over the growing season when growing rust-resistant cultivars (mid-July through October). Roughly 30% of farmers applied urea as an N fertilizer to soils shortly before harvest (October to November).

At the time of sampling (i.e., dry season), most soil extractable and total leaf nutrient

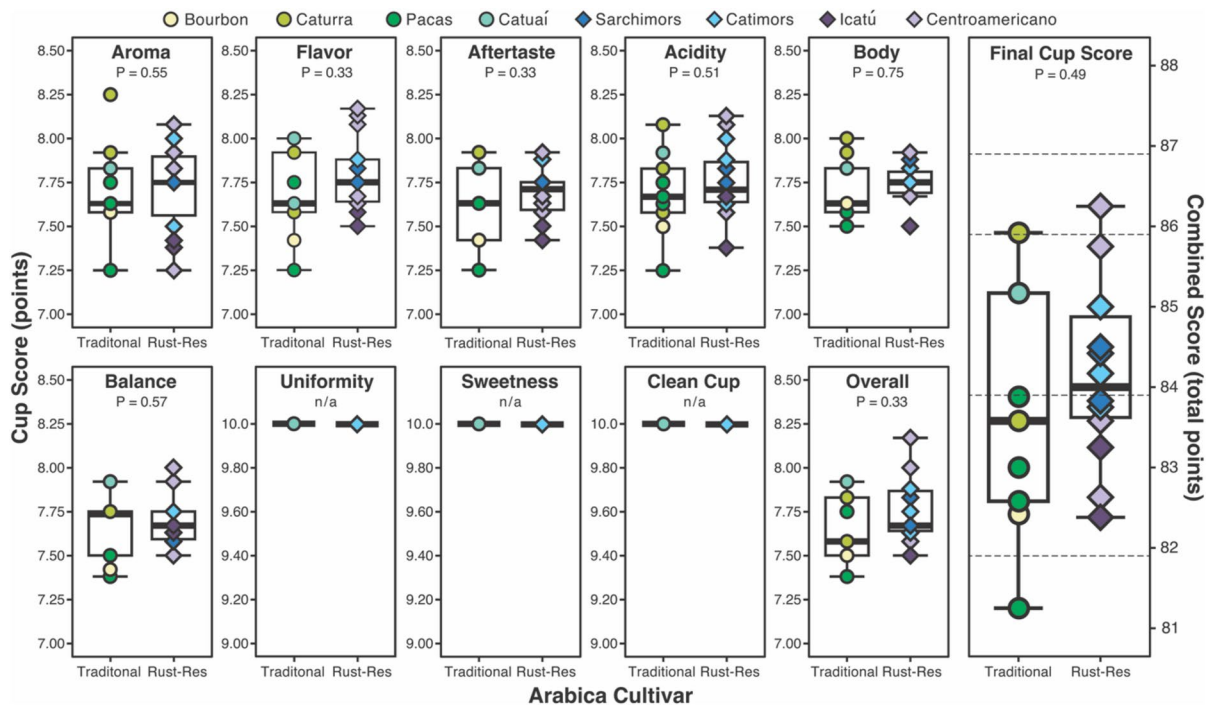


Fig. 1 Coffee cup qualities and final cup scores assigned to Arabica coffee (*Coffea arabica* L.) grown in fields in south-central Guatemala ($n=23$ total fields). Fields were dominated by cultivars that ascended from **A** Bourbon and Typica lineages traditionally grown in Central America (Bourbon, Caturra, Pacas, Catuai; $n=9$ fields), or **B** rust-resistant cultivars derived from Arabica $\times$ *C. canephora* progenies (Sarchimors, Catimors, Icatú, Centroamericano; $n=14$ fields). Symbols median scores assigned by two or three Coffee Quality

Institute (CQI) certified Arabica coffee Q-graders (SCA 2020b) and p-values correspond with results from a Wilcoxon rank-sum test. Horizontal dashed lines on right-hand panels correspond with scores used differentiate specialty coffee free on board (pre-export) prices (Roberts and Treweek 2024). n/a=not applicable due to all CQI graders assigning perfect scores for uniformity, sweetness, and clean cup qualities (i.e., 10.0 points)

concentrations varied across fields (Table 2) and minimally between cultivars (Table S5 and S6). Soils in $>85\%$ of fields had ≈ 2 - to 3.5-fold lower $\text{NO}_3\text{-N}$ (mean = 6.0 mg kg^{-1}) and K (mean = 57.3 mg kg^{-1}) concentrations than minimum thresholds generally considered ideal for crop production. Soil $\text{NH}_4\text{-N}$ (mean = 3.3 mg kg^{-1}), Mg (mean = 80.4 mg kg^{-1}) and B (mean = 0.28 mg kg^{-1}) concentrations were 1.3- to 1.8-fold lower than minimum thresholds in 70–100% of fields. Soil Al concentrations in all fields were an order of magnitude higher than other soil extractable nutrients (except Ca). Leaf nutrient concentrations in $>75\%$ of fields were 1.2- to 1.8-fold lower for S (mean = 0.17%), B (mean = 48 mg kg^{-1}), Zn (mean = 9.0 mg kg^{-1}), and Cu (mean = 9.0 mg kg^{-1}) than minimum thresholds recommended for Central American Arabica coffee production. Nutrient concentrations for green beans were also largely similar

between cultivars, other than marginally higher Mg concentrations in traditional versus rust-resistant cultivars ($p=0.07$) (Table S6).

Relationships between cup quality and measures of soil fertility and plant nutrition

Traditional cultivars exhibited fewer univariate relationships with soil fertility measures and leaf and green bean nutrients—opposite of relationships for rust-resistant cultivars (Fig. 3). Traditional cultivar body and balance scores were negatively correlated with soil CEC ($\text{PCC} \approx -0.70$), whereas all cup qualities (except aftertaste) and final cup scores for rust-resistant cultivars were positively correlated with soil K concentrations ($\text{PCC}=0.62$ to 0.80) (Fig. 3A). Rust-resistant cultivars also exhibited positive relationships between aroma scores and soil Cu and Al

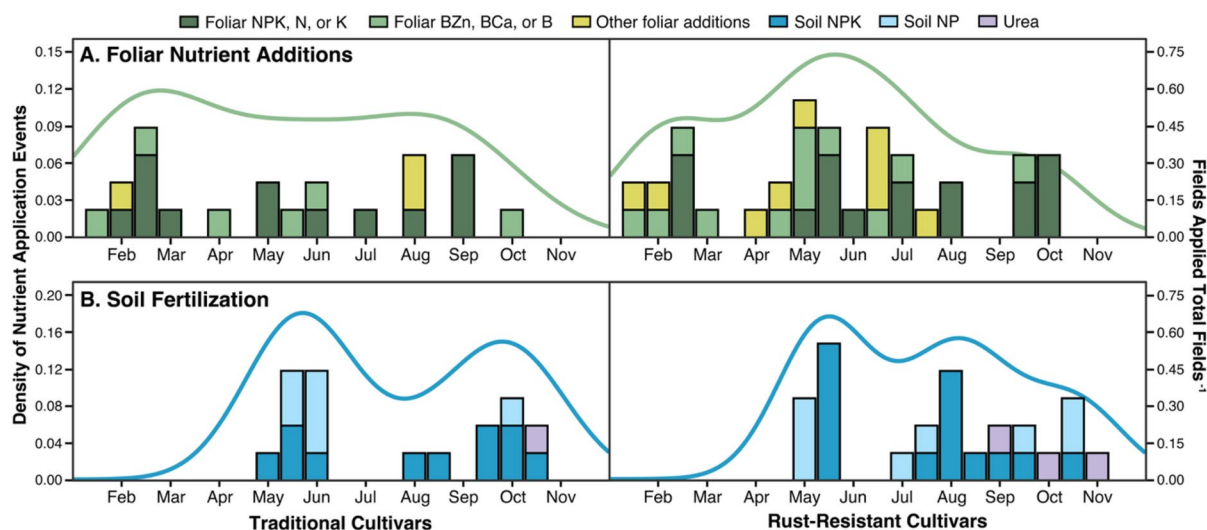


Fig. 2 Frequency and temporal distribution of **A** foliar nutrient application and **B** soil fertilization in Arabica coffee (*Coffea arabica* L.) fields in south-central in Guatemala ($n=23$ total fields). Fields were dominated by cultivars that ascended from (i) Bourbon and Typica lineages traditionally grown in Central America (Bourbon, Caturra, Pacas, Catuái; $n=9$ fields), or (ii) rust-resistant cultivars derived from Arabica $\times$ *C. canephora* progenies (Sarchimors, Catimors, Icatú,

Centroamericano; $n=14$ fields). Foliar nutrients were applied in nitrogen-phosphorus-potassium (NPK) and boron (B)-zinc (BZn), or B-calcium (BCa) mixtures, or as single nutrient N, K, or B solutions and proprietary and/or locally fermented blends of unknown compositions (other). Soil fertilizer was applied in NPK (proportions of 20-20-20, 19-14-19, 18-6-12, 16-4-16, or 15-15-15), NP (NPK proportions of 20-20-0), or as urea-N

concentrations ($PCC \approx 0.60$), and balance and overall scores and soil Mg concentrations ($PCC=0.54$). Conversely, nearly all traditional cultivar cup qualities and final cup scores were positively correlated with leaf Zn concentrations ($PCC=0.52$ to 0.90) and negatively correlated with green bean S ($PCC=-0.36$ to -0.67) and B concentrations ($PCC=-0.40$ to -0.54), whereas all cup qualities (except aroma) and final cup scores for rust-resistant cultivars were negatively correlated with leaf K concentrations ($PCC=-0.57$ to 0.81) (Fig. 3B). Traditional cultivars also exhibited negative relationships between green bean N concentrations and flavor, aftertaste, acidity, and final cup scores ($PCC=-0.29$ to -0.50), and rust-resistant cultivars exhibited positive relationships between green bean Mn concentrations and acidity scores ($PCC=0.50$) and green bean Fe ($PCC=0.60$) and Al concentrations ($PCC=0.54$) and aftertaste scores.

Most cup qualities (6 of 7) and final cup scores for traditional cultivars were best predicted by total leaf and green bean nutrient concentrations (Fig. 4A), whereas most rust-resistant cultivar cup qualities (5 of 7) were best predicted by soil fertility

measures (Fig. 4B). Univariate and multivariate LASSO model estimates were reasonably accurate overall ($R^2_{adj.}=0.32$ – 0.82 , $RMSE=0.10$ – 0.23 points), though with some notable cultivar-specific differences (Table 3). For instance, aftertaste scores for traditional cultivars were more accurately predicted by leaf Zn (55% importance), green bean S (45% importance), and soil K (<1% importance) ($R^2_{adj.}=0.82$) than by leaf K, leaf Zn, and green bean Fe for rust-resistant cultivars ($R^2_{adj.}=0.66$). Conversely, body scores more accurately predicted by leaf K concentrations for rust-resistant cultivars ($R^2_{adj.}=0.62$) compared to soil CEC for traditional cultivars ($R^2_{adj.}=0.43$). Notably, most cup qualities (5 of 7) for traditional cultivars were best predicted by total leaf Zn (55–100% importance), which was also the single best predictor of traditional cultivar final cup score ($R^2_{adj.}=0.43$). By comparison, all cup qualities for rust-resistant cultivars were best predicted by soil extractable and/or total leaf K concentrations (25–100% importance) as were final cup scores ($R^2_{adj.}=0.73$). Overall, models were less accurate across cup qualities for Pacas and Caturra (30% of total residual variance) (Fig. 5A)

Table 2 Summary statistics soil fertility (0–15 cm depth) and plant nutrition (as proxied by total leaf nutrient concentrations) for Arabica coffee (*Coffea arabica* L.) grown in fields in south-central Guatemala (n=23 fields). Fields were dominated by cultivars that ascended from (i) Bourbon and Typica lineages traditionally grown in Central America (Bourbon, Caturra, Pacas, Catuai; n=9 fields), or (ii) rust-resistant cultivars derived from Arabica × *C. canephora* progenies (Sarchimors,

Catimors, Icatú, Centroamericano; n=14 fields). Percentages reported below minimum thresholds indicate the proportion fields (total and fields planted to specific cultivars) with potential nutrient deficiencies based on nutrient concentrations in soils generally considered ideal for crop production (Wang et al. 2015) and leaf nutrient concentrations recommended for Arabica coffee production in Central America (Llatch-Hill 2017)

Measures	Minimum threshold	Summary statistics					Sites below thresholds	
		Min	Mean	Median	Max	CV	Traditional	Rust-resistant
<i>Soil fertility^a</i>								
pH	6	5.9	6.3	6.4	6.6	3%	11% (n=1)	7% (n=1)
CEC (m _{eq} 100 g ⁻¹)	8	6.6	9.3	9.1	13.8	20%	22% (n=2)	29% (n=4)
SOC (%)	0.5	1.6	2.3	2.4	3.6	23%	0% (n=0)	0% (n=0)
TSN (%)	n/a	0.15	0.23	0.23	0.37	24%	–	–
C:N	n/a	9.4	10.2	10.2	11.5	6%	–	–
NH ₄ ⁺ -N (mg kg ⁻¹)	5	2.0	3.4	3.3	5.3	29%	89% (n=8)	93% (n=13)
NO ₃ ⁻ -N (mg kg ⁻¹)	20	1.3	6.0	4.6	18.5	71%	100% (n=9)	100% (n=14)
P (mg kg ⁻¹)	25	15.0	63.3	56.0	186.0	62%	11% (n=1)	7% (n=1)
K (mg kg ⁻¹)	110	23.0	57.3	45.0	135.0	53%	89% (n=8)	93% (n=13)
S (mg kg ⁻¹)	5	6.0	7.2	7.0	10.0	13%	0% (n=0)	0% (n=0)
Ca (mg kg ⁻¹)	150	540	1164	1124	1794	28%	0% (n=0)	0% (n=0)
Mg (mg kg ⁻¹)	110	39.0	80.4	74.0	133.0	36%	89% (n=8)	71% (n=10)
B (mg kg ⁻¹)	0.5	0.01	0.28	0.28	0.55	43%	100% (n=9)	93% (n=13)
Zn (mg kg ⁻¹)	1.0	2.82	8.26	6.61	35.55	89%	0% (n=0)	0% (n=0)
Cu (mg kg ⁻¹)	0.5	2.57	5.56	5.71	6.98	21%	0% (n=0)	0% (n=0)
Mn (mg kg ⁻¹)	1.0	2.0	5.3	5.0	9.0	34%	0% (n=0)	0% (n=0)
Fe (mg kg ⁻¹)	50.0	143	173	173	211	11%	0% (n=0)	0% (n=0)
Al (mg kg ⁻¹)	n/a	908	1437	1455	1720	16%	–	–
<i>Plant nutrition</i>								
Leaf N (%)	2.30	2.22	2.68	2.65	3.33	9%	0% (n=0)	7% (n=1)
Leaf P (%)	0.12	0.10	0.15	0.15	0.18	16%	22% (n=2)	14% (n=2)
Leaf K (%)	1.70	0.17	0.33	2.45	0.53	34%	11% (n=1)	7% (n=1)
Leaf S (%)	0.20	0.14	0.17	0.17	0.19	8%	89% (n=9)	93% (n=13)
Leaf Ca (%)	1.00	1.25	1.69	1.66	2.29	17%	0% (n=0)	0% (n=0)
Leaf Mg (%)	0.20	0.17	0.33	0.29	0.53	34%	22% (n=2)	0% (n=0)
Leaf B (mg kg ⁻¹)	60	26.1	48.0	45.7	74.0	26%	78% (n=7)	86% (n=12)
Leaf Zn (mg kg ⁻¹)	15	6.7	9.0	8.7	17.2	24%	100% (n=9)	93% (n=13)
Leaf Cu (mg kg ⁻¹)	12	3.3	7.0	6.3	15.1	44%	100% (n=9)	86% (n=12)
Leaf Mn (mg kg ⁻¹)	50	17.9	46.7	39.9	85.7	47%	67% (n=6)	57% (n=18)
Leaf Fe (mg kg ⁻¹)	75	340	463	453	646	19%	0% (n=0)	0% (n=0)
Leaf Al (mg kg ⁻¹)	n/a	349	638	629	897	22%	–	–

CEC, cation exchange capacity; SOC, soil organic carbon; TSN, total soil nitrogen; C:N, carbon-to-nitrogen ratio; NH₄⁺-N, ammonium-N; NO₃⁻-N, nitrate-N; P, phosphorus; K, potassium; S, sulfur; Ca, calcium; Mg, magnesium; B, boron; Zn, zinc; Cu, copper; Mn, manganese; Fe, iron; Al, aluminum; n/a, not applicable (i.e., no specified threshold)

^aExtractable soil nutrient concentrations were 0.5 M K₂SO₄ extractable soil NO₃⁻-N and NH₄⁺-N, and Mehlich-3 extractable P, K, Ca, Mg, B, Zn, Cu, Mn, Fe, and Al

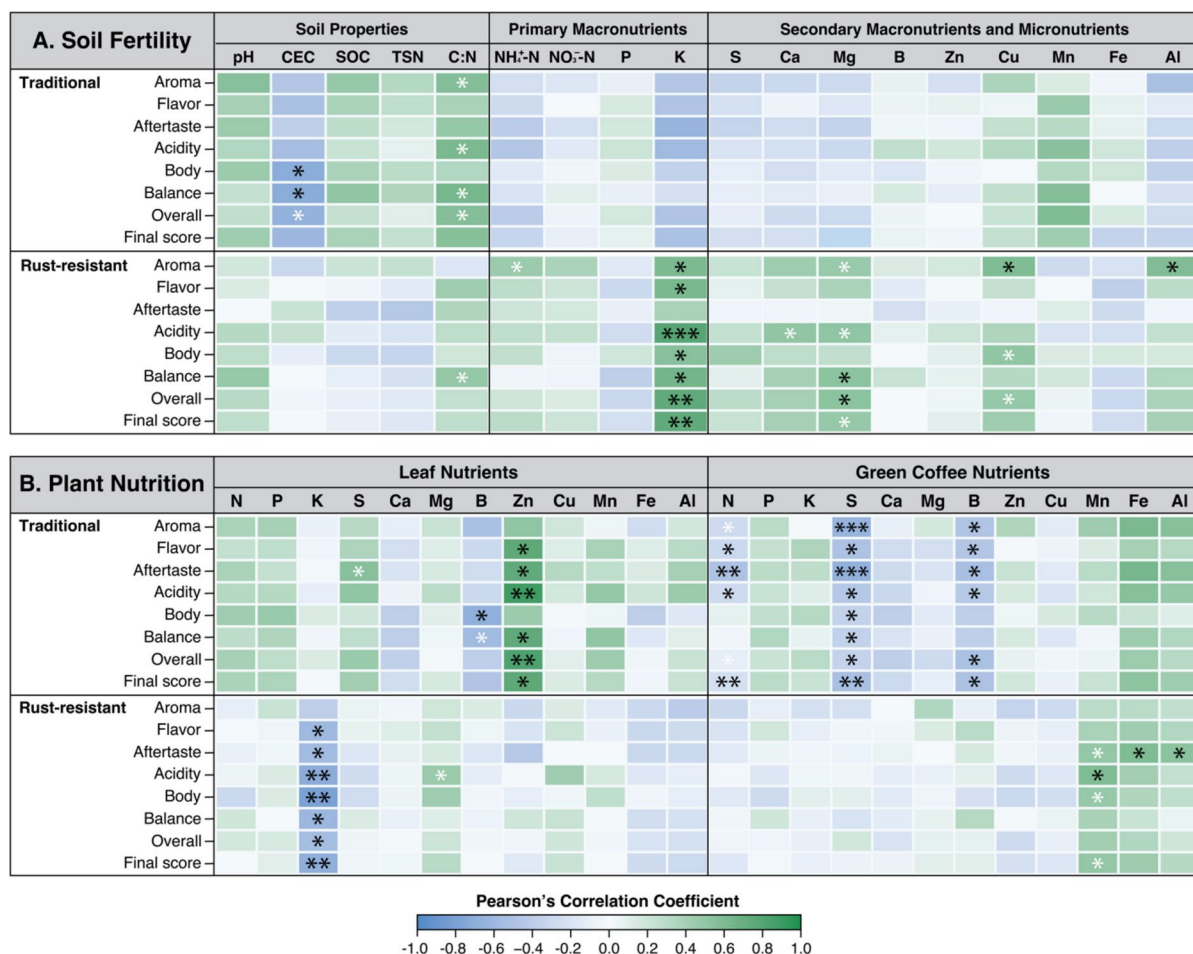


Fig. 3 Univariate relationships (Pearson's correlation coefficients [PCC]) between cup qualities and final cup scores and measures of **A** soil fertility (0–15 cm depth) and **B** plant nutrition as (leaf and green bean nutrients) for Arabica coffee (*Coffea arabica* L.) grown in fields (n=23) in south-central Guatemala. Fields were dominated by cultivars ascended from (i.) Bourbon and Typica lineages traditionally grown in Central America (Bourbon, Caturra, Pacas, Catuai; n=9 fields), or (ii.) rust-resistant cultivars derived from Arabica×*C. canephora* progenies (Sarchimors, Catimors, Icatú, Centroamericano; n=14 fields). Mean cup scores were assigned by two or three Coffee Quality Institute (CQI) certified Arabica coffee Q-grad-

ers (SCA 2020b). Extractable soil nutrient concentrations were 0.5 M K₂SO₄ extractable soil nitrate-N (NO₃⁻-N) and ammonium-N (NH₄⁺-N), and Mehlich-3 extractable phosphorus (P), potassium (K), sulfur (S), calcium (Ca), magnesium (Mg), boron B, zinc (Zn), copper (Cu), manganese (Mn), iron (Fe), and aluminum (Al). Cation exchange capacity is the summation Mehlich-3 extractable K, Ca, Mg, and sodium. Asterisks denote the level of significance (white= $p \leq 0.10$; black= $p \leq 0.05^*$, $p \leq 0.01^{**}$, $p \leq 0.001^{***}$). Uniformity, sweetness, and clean cup qualities received perfect scores from all CQI graders (i.e., 10.0 points) and thus were not included in our analysis nor this figure (see Fig. 2)

and Centroamericano (35% total residual variance) (Fig. 5B). Across Arabica cultivars, models over-estimated final cup scores associated with higher free onboard specialty market prices (> 85.0 points) compared to lower but reasonable prices (82.0–84.9 points) (Fig. 5C).

Discussion

Arabica genetics minimally impacts coffee cup scores in this region

We find that traditional and rust-resistant Arabica cultivars grown in this region of Guatemala are similar

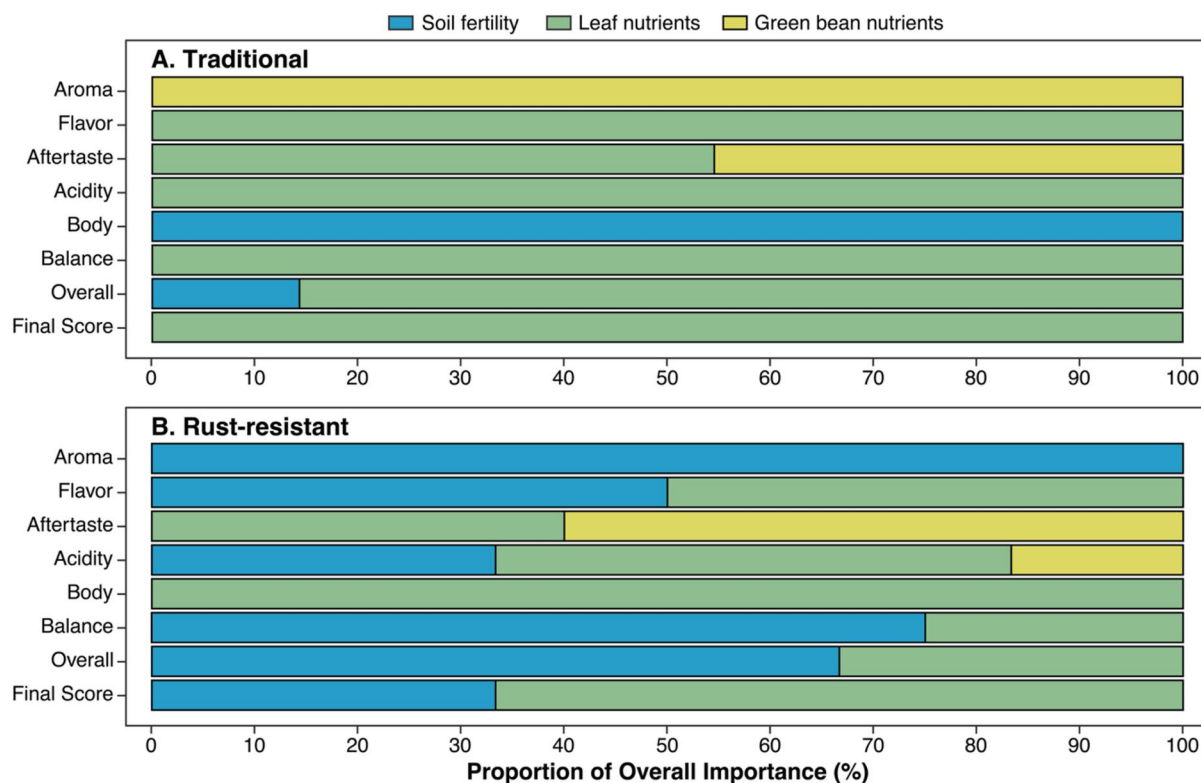


Fig. 4 Comparative proportions of the overall importance of soil fertility (0–15 cm depth) versus plant nutrition (leaf and green bean nutrients) measures in predicting cup qualities and final cup scores for Arabica coffee (*Coffea arabica* L.) grown in fields in south-central Guatemala dominated by cultivars ascended from **A** Bourbon and Typica lineages traditionally grown in Central America (Bourbon, Caturra, Pacas, Catuái; $n=9$ fields), **B** rust-resistant cultivars derived from Arabica $\times$ *C. canephora* progenies (Sarchimors, Catimors, Icatú, Centroamericano; $n=14$ fields). Bars represent the aggregated relative importance of soil fertility versus plant nutri-

tion variables in predicting coffee cup scores based on univariate regression (i.e., 100% importance) or (ii) least absolute shrinkage and selection operator (LASSO) regression models (i.e., individual standardized β coefficients relative to their absolute sum) predicted median cup scores assigned by two or three Coffee Quality Institute (CQI) certified Arabica coffee Q-graders (SCA 2020b). Uniformity, sweetness, and clean cup qualities received perfect scores from all CQI graders (i.e., 10.0 points) and thus were not included in our analysis nor this figure (see Fig. 2)

in cup quality, contrary to our hypothesis and the proposed rust resistance-cup quality trade-offs of cultivars carrying genomic sequences introgressed from *C. canephora* (Toniutti et al. 2017, 2019). Appreciable variation in final cup scores for traditional cultivars could entail unstable specialty coffee market access given consumers are willingly pay higher prices for unique flavor profiles and consistently superior cup qualities (Grabs and Ponte 2019; Bressani et al. 2021). However, because the 100-point Arabica coffee Q-grading system does not directly assess extrinsic impacts on cup quality (e.g., origin, post-harvest processing) nor account for consumer preferences (Sörqvist et al. 2013; Giacalone et al. 2016;

Samoggia and Riedel 2018), cup score does not necessarily reflect current specialty coffee market trends. Thus, blanket recommendations to expand specialty coffee market access by replacing traditional Arabicas with rust-resistant cultivars first requires more extensive environment $\times$ management $\times$ market-fit access evaluations.

Coffee cup quality similarities in observed in this study may reflect farmers selecting cultivars tailored to field-specific characteristics and growing conditions. For example, the lower average elevation of rust-resistant (1240 ± 43 m asl) versus traditional (1341 ± 39 m asl) cultivars in our study region may reflect Arabica coffees containing *C. canephora* genes

Table 3 Best-fit linear regression models predicting cup qualities and final cup scores for Arabica coffees (*Coffea arabica* L.) from fields in south-central Guatemala that were dominated by cultivars that ascended from (i) Bourbon and Typica lineages traditionally grown in Central America (Bourbon, Caturra, Pacas, Catuai; n=9 fields), (ii) rust-resistant cultivars derived from Arabica×*C. canephora* progenies (Sarchimors, Catimors, Icatú, Centroamericano; n=14 fields). Median scores assigned by two or three Coffee Quality Institute (CQI) certified Arabica coffee Q-graders (SCA 2020b) that were pre-

dicted by measures of soil fertility (0–15 cm depth) and plant nutrition (leaf and green bean nutrients) using univariate and multivariate least absolute shrinkage and selection operator (LASSO) regression. Values in parentheses correspond with estimated univariate model coefficients or standardized β coefficients and their relative importance (i.e., individual standardized β coefficients relative to their absolute sum). Uniformity, sweetness, and clean cup qualities received perfect scores from all CQI graders (i.e., 10.0 points) and thus were not included in our analysis nor this table (see Fig. 2)

Cultivar	Cup quality	Soil fertility	Plant nutrition		Model accuracy	
			Leaves	Green beans	R ² _{adj}	RMSE
Traditional	Aroma	–	–	S (–0.24)	0.32	0.23 pts
	Flavor	–	Zn (0.04)	–	0.53	0.21 pts
	Aftertaste	K (<0.01, 1%)	Zn (0.06, 55%)	S (–0.05, 45%)	0.82	0.15 pts
	Acidity	–	Zn (0.19)	–	0.77	0.10 pts
	Body	CEC (–0.08)	–	–	0.43	0.13 pts
	Balance	–	Zn (0.14)	–	0.52	0.13 pts
	Overall	CEC (–0.01, 14%)	Zn (0.06, 86%)	–	0.75	0.13 pts
	Final score	–	Zn (0.09)	–	0.53	1.33 pts
Rust-resistant	Aroma	K (0.04)	–	–	0.34	0.19 pts
	Flavor	K (0.01, 50%)	K (–0.01, 50%)	–	0.47	0.18 pts
	Aftertaste	–	K (–0.02, 40%) Zn (–0.01, 20%)	Fe (0.02, 40%)	0.66	0.11 pts
	Acidity	K (0.02, 33%)	K (–0.03, 50%)	Mn (0.01, 17%)	0.78	0.11 pts
	Body	–	K (–0.03)	–	0.62	0.11 pts
	Balance	C:N (0.02, 50%) K (0.01, 25%)	K (–0.01, 25%)	–	0.56	0.11 pts
	Overall	K (0.01, 33%)	K (–0.01, 33%)	–	0.56	0.14 pts
	Final score	K (0.01, 33%)	K (–0.02, 67%)	Fe (<0.01, <1%)	0.73	0.63 pts

R²_{adj}, adjusted correlation coefficient; RMSE, root-mean square error; S, sulfur; Zn, zinc; K, potassium; CEC, cation exchange capacity; Fe, iron; Mn, manganese; C:N, carbon-to-nitrogen ratio

being better suited to warm air temperatures at lower elevations compared to Bourbon and Typica lineages that thrive exclusively in high-elevation environments (DaMatta et al. 2007). High concentrations of non-marketable, small (<1 mm) or physically defected beans and low density cherries (i.e., cherries) unseasonably elevated air temperatures (>32 °C) and/or low soil matric potentials (<–8 MPa) may have accelerated cherry maturation in traditional cultivars to a greater extent than rust-resistant cultivars with greater environmental stability (Table S4) (Vaast et al. 2006; DaMatta et al. 2012; Pereira et al. 2016). Although final cup scores suggest growing conditions the season our study was conducted may have been favorable for traditional cultivars known for their environmental sensitivity (van der Vossen et al.

2015), the lack of proximal weather monitoring stations definitively linking physical and cup quality attributes to cultivar-specific environmental sensitivity, a challenge in much of the global Coffee Belt.

Our results partially support the emerging body research that the F₁ hybrid Centroamericano can deliver higher cup quality than traditional cultivars across variable conditions. Centroamericano obtained the highest final cup score (86.25 points) and represented substantial proportion (≈35%) of cultivars in this study that would have fetched a high free on-board specialty market price (>84.0 points) (Roberts and Trewick 2024). A similar study found that Centroamericano obtained higher cup scores and cherry yields than traditional cultivars over a 500-m elevational gradient in Central America (Marie et al.

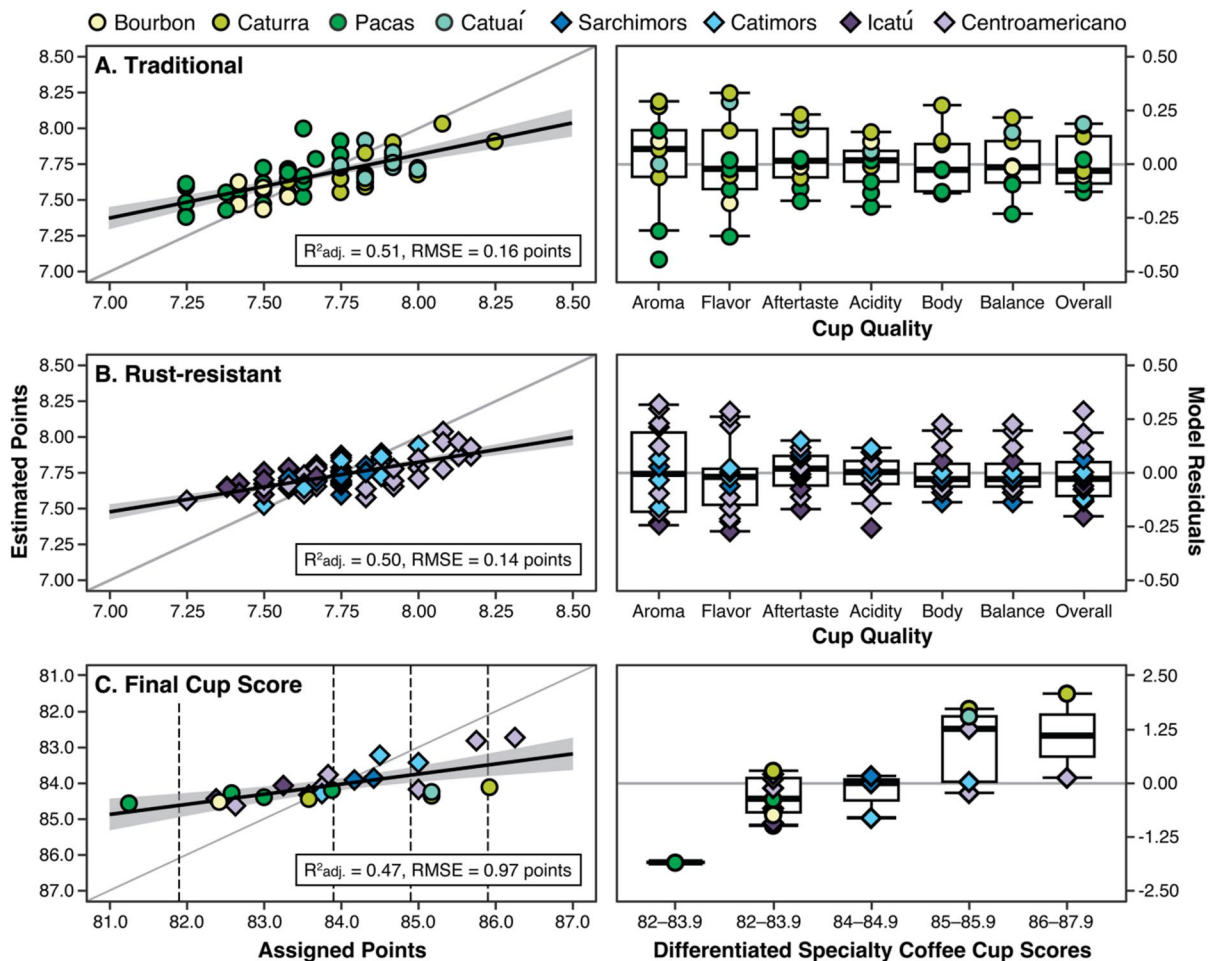


Fig. 5 Accuracy assessment for best-fit linear models predicting Arabica coffee (*Coffea arabica* L.) cup qualities for cultivars that ascended from **A** Bourbon and Typica lineages traditionally grown in Central America (Bourbon, Caturra, Pacas, Catuai; $n=9$ fields), **B** rust-resistant cultivars derived from Arabica $\times$ *C. canephora* progenies (Sarchimors, Catimors, Icatú, Centroamericano; $n=14$ fields) and **C** final cup scores across cultivars and point intervals (vertical dashed lines) used to differentiate specialty coffee free on board (pre-export) prices (Roberts and Treweek 2024). Symbols are median scores assigned by two or three Coffee Quality Institute (CQI) certi-

fied Arabica coffee Q-graders (SCA 2020b) that were predicted by soil fertility (0–15 cm depth) plant nutrition (total leaf and green (i.e., unroasted) coffee nutrients) measures using univariate and multivariate least absolute shrinkage and selection operator (LASSO) regression (Table 3). Adjusted correlation coefficients ($R^2_{adj.}$) and root-mean square errors (RMSE) correspond with regression lines and ribbons (95% confidence intervals) as indicators of model accuracy. Uniformity, sweetness, and clean cup qualities received perfect scores from all CQI graders (i.e., 10.0 points) and thus were not included in our analysis (see Fig. 2)

2020). Centroamericano yields are generally higher and more stable than traditional cultivars across variable conditions of shade coverage (Bertrand et al. 2011) and fertilizer inputs (Allen Asensio et al. 2026). Although this points to the agronomic potential of F_1 hybrids at high-elevation under shaded tree canopies and intensive production under full sun (Bertrand et al. 2026) emerging evidence suggest F_1

hybrids may incur higher labor (90%) and fertilizer (48%) costs to meet production goals (Kahsay et al. 2023) and are inconsistently sensitive to variable soil moisture conditions (Pappo et al. 2021; Sarzynski et al. 2024). Thus, the potential cup quality and yield benefits afforded by F_1 hybrids may not be sufficiently reliable to warrant replacing traditional cultivars in all management and environmental contexts.

Furthermore, because F_1 hybrids are currently propagated clonally via somatic embryogenesis rather than by seed (Bertrand et al. 2026), the per plant cost to obtain F_1 hybrids is substantially higher (\$0.75 USD per plantlet) than seed-based progenies (\$0.25 USD per seedling) (WCR 2024). The only seed-based F_1 hybrids currently commercially available are Starmaya (Georget et al. 2019) and Mariana more recently (Bertrand et al. 2026), but newly identified male-sterile sources from Latin America (Suárez and Flórez Ramos 2023) are may expand commercial F_1 hybrid seed production once validated (Bertrand et al. 2026).

Soil fertility does not inform coffee plant nutrition under current management practices

We found potentially soil nutrient deficiencies rarely corresponded with leaf concentrations sub-optimal for Arabica coffee production in Central America, regardless of cultivar-specific variation in the timing and formulation of soil and foliar nutrient additions. For example, soil inorganic N and K concentrations across coffee fields were below minimum thresholds generally considered ideal for crop production, despite the dominance of soil NPK fertilizers applied by farmers in the region. Conversely, Zn deficiencies were more commonly observed in coffee leaves than in soils despite repeated co-application of foliar Zn and B mixtures—all fields had B deficient soils and leaves. These findings suggest soil fertility measures alone may not adequately reflect the nutritional status of mature Arabica trees, which is consistent with smallholder Robusta coffee production systems in South America (Kasmerchak et al. 2025). Although we cannot identify the exact mechanism(s) underpinning leaf vs soil nutrient discrepancies in our observational study, it is plausible that sampling at peak-harvest did not capture wet season nutrient status when total rainfall was nearly two-fold higher (4015 mm) compared the dry season (2382 mm) in the 2019/20 crop year when sampling occurred.

Our findings suggest that at rates currently used by farmers, applying NP and NPK fertilizer once before the wet season (May to June) does not appreciably impact mature Arabica tree nutritional status throughout the growing season. High total rainfall throughout the wet season (4015 mm) and low soil CEC ($6.0 \text{ meq } 100 \text{ g}^{-1}$) favors fertilizer leaching in coarse

textured Andisols with low nutrient holding capacity (Allen Asensio et al. 2024). Negligible to negative agroeconomic N and K use efficiencies from on-farm coffee fertilization trials on these same fields (Allen Asensio et al. 2026) and generally low N fertilizer persistence (e.g., only 50% ^{15}N -fertilizer recovery as soil NO_3^- -N at 7 weeks after application) (Cannavo et al. 2013) further suggest fertilizer efficiency for coffee may be limited in steeply sloped, humid volcanic landscapes. This may in part explain limited use of fertilizers on an absolute basis (kg ha^{-1}) on smallholder farm throughout the volcanic landscapes in the Central American Coffee Belt compared to intensified coffee plantations (Hagggar et al. 2011; Harvey et al. 2021), which in turn could reflect non-nutrient yield limitations for coffee (e.g., pathogens, sunlight) (Wang et al. 2015; Bhattarai et al. 2017).

Peak harvest soil vs coffee leaf nutrient discrepancies may also partially reflect differences in locally-tailored foliar nutrient management practices. Repeated application of foliar N and/or K is a low-cost, rapid approach to correct severe N and K deficiencies under conditions favoring fertilizer leaching losses (Fageria et al. 2009; Tay 2022) and minimize impacts to production from coffee leaf rust and American leaf spot (*Mycena citricolor*) infection (Montero et al. 2018; Brown 2019). Unlike soil inorganic N and K deficiencies, widespread soil and leaf B deficiencies across fields in our study likely reflect the high B adsorption capacity of abundant short-range order minerals (e.g., allophane) in volcanic soils in this landscape (Terraza Pira et al. 2018) and generally low B mobility in phloem sap (Hansch and Mendel 2009; Silva et al. 2021; Rengel et al. 2022). Most farmers (50%) co-applied foliar B with Zn at the start of the rainy season (April to May) or immediately after cherry harvest ($\approx 35\%$), which suggests our sampling at peak-harvest may not have captured translocation of micronutrients from leaves to flowers during a transient period of post-harvest plant stress (Silva et al. 2021) or lower leaf adsorption of foliar B and Zn additions during the dry season (Clemente et al. 2018).

Inferring coffee cup quality from soil and plant nutrients: reason for caution

Cultivar-specific relationships between cup scores and soil fertility and plant nutrition measures identified in our study suggest soil and leaf nutrient concentrations often considered deficient may not

substantially diminish coffee cup quality, contrary to past assertions (Mitchell 1988; Feria-Morales 2002; Läderach 2007). Limited soil nutrient availability and the resulting coffee plant nutrient deficiencies should also hypothetically modulate cup quality (Morales-Ramos et al. 2020) but coffee cup scores were consistently high (>81.2 points) across variable management practices and potential soil and leaf nutrient deficiencies. Building on past assessments (e.g., Bertrand et al. 2006, 2012; Marie et al. 2020), we find that the impacts of Arabica genetics (G) on coffee cup quality are conditioned by environmental (E) and management (M) factors (i.e., $G \times E \times M$). However, establishing relationships between cup scores and measures of soil fertility and/or plant nutrition larger network of sites is needed to identify reliable indicators of coffee cup quality across variable soil properties and management conditions in the Central American Coffee Belt.

Cultivar-specific relationships between cup scores and soil extractable and total leaf nutrient concentrations demonstrates the role of plant genetics in modulating coffee cup quality in this landscape. The positive relationship traditional cultivar cup qualities and leaf Zn may reflect increased enzymatic oxidation of phenols associated with bitter and astringent flavors (Lacerda et al. 2018) and enhanced photosynthesis (Clemente et al. 2018) with increasing Zn, and could reflect leaf concentrations being indicative of plant Zn deficiency. Similarly, ≈ 1.5 - to 2-fold lower soil extractable K concentrations than minimum thresholds should have corresponded with lower cup scores in rust-resistant cultivars, but Centroamericano obtained the highest cup score (86.25 points). Other soil properties (e.g., microbial biomass) and/or edaphic factors (e.g., light availability) not examined in our study could also be stronger determinants of coffee cup quality than soil extractable and total leaf nutrients, which warrants expanding evaluations across soil properties and climatic conditions over a larger network of sites. Notably, Central American farmers in our study area (Brown 2019) and other regions (Turreira-García 2022) have reported minimal management guidance for F_1 hybrids. Contrary to the common perception that rust-resistant cultivars are inferior to traditional cultivars with respect to cup quality (Tay 2025), we find that Arabica cultivars with rust-resistant genes can have similarly appealing

organoleptic cup qualities as traditional cultivars regarded for their high cup quality (Fig. 6).

We recognize our harvesting approach may not necessarily represent the cup quality Arabica grown under regional management practices, as we sorted and processed individual cherries based on optimal quality attributes (i.e., ripeness, minimal external defects). This may partially explained markedly higher final coffee cup scores in our study compared to traditional (76.3 points) and rust-resistant cultivars (80.5 points) grown at comparable elevations in Nicaragua (710–1250 m asl) (Marie et al. 2020). However, the SCA standard Arabica Q-grading system assesses cup quality based on objective human perceptions subjected to the accuracy of individual graders (Marcus et al. 2023). Although CQI certified Q-graders in our study assessed three cups per coffee rather than five cups used in the standard SCA protocol (SCA 2020a), assigned cup scores for individual CQI graders varied from median scores by 2–5% in most cases (Table S7) and we intentionally analyzed median cup scores to better constrain grader subjectivity. It is worth noting that human subjectivity and other extrinsic factors (e.g., origin, post-harvest processing) are evaluated in the SCA's new industry standard Coffee Value Assessment (CVA) framework (SCA 2024), which replaced the 100-point Q-grading system in 2024 as the market standard after cup quality was assessed for the present study. Thus, potential bias introduced by selective cherry harvesting practices and subjective human perception would still be expected for any sensory value-based assessment of coffee cup quality.

Conclusions

Identifying the impacts of soil fertility and plant nutrition on *C. arabica* cup quality across on-farm sites is needed to evaluate the suitability of rust-resistant Arabica cultivars compared to traditional American regarded for high cup quality. Contrary to our hypothesis, traditional cultivars were not superior to rust-resistant cultivars with respect to cup score, regardless of appreciable within-cultivar variation in management practices and soil extractable and total leaf nutrient concentrations. Soil inorganic N, K, Mg, B concentrations were below minimum thresholds generally considered optimal for crop

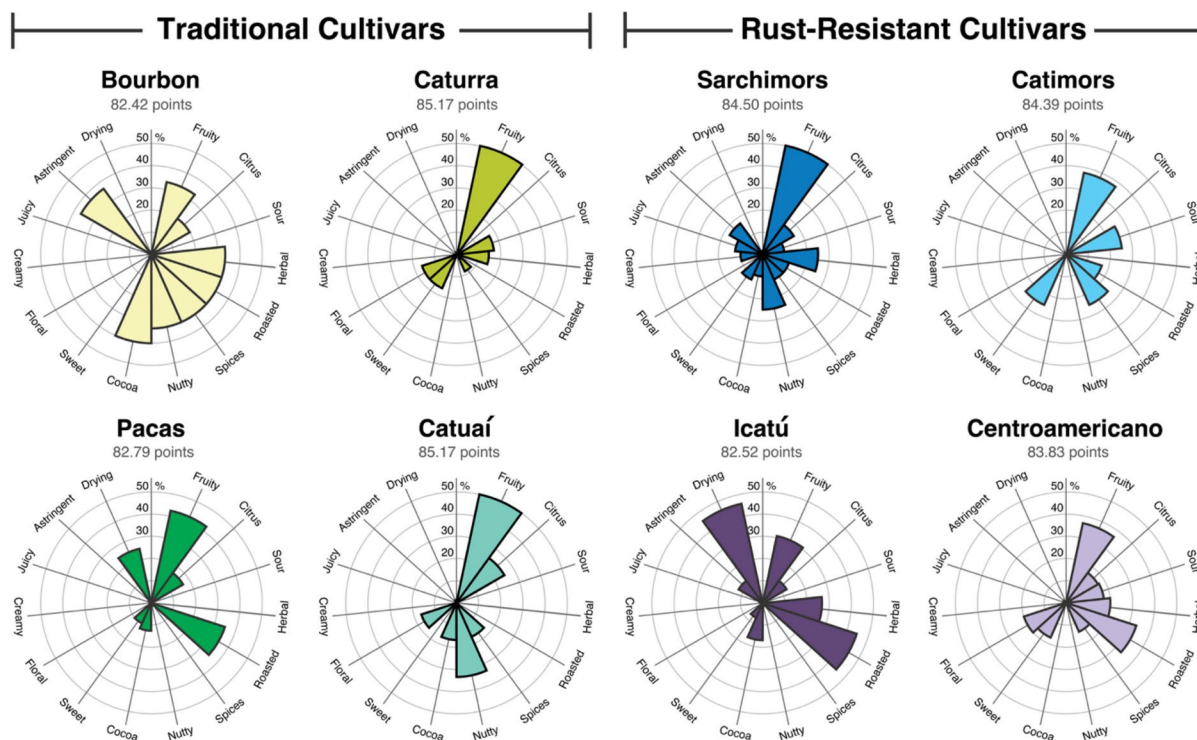


Fig. 6 Wind-rose diagrams comparing the median relative (%) occurrence of coffee cup flavor profiles and tasting notes for Arabica coffee (*Coffea arabica* L.) grown in fields in south-central Guatemala ($n=23$ fields). Fields were dominated by cultivars that ascended (i.) Bourbon and Typica lineages traditionally grown in Central America (Bourbon, Caturra, Pacas, Catuai; $n=9$ fields), (ii.) rust-resistant cultivars derived from Arabica $\times$ *C. canephora* progenies (Sarchimors, Catimors, Icatú, Centroamericano; $n=14$ fields). Following SCA proto-

cols (SCA 2020b), flavor profiles and tasting notes were characterized by two or three Coffee Quality Institute (CQI) certified Arabica coffee Q-graders using terms from the World Coffee Research (WCR) Sensory Lexicon (WRC 2024), which were then aggregated and compared based on specific cup quality attributes. Bars represent median occurrences across CQI graders within cultivars ($n=1-7$ fields per cultivar). Median final cup scores are displayed above each diagram for context

production in most fields, as were leaf S, Zn, Cu, and B concentrations considered optimal for Arabica production in Central America. Most cup qualities and final cup scores were correlated with leaf Zn concentrations for traditional cultivars and soil and/or leaf K concentrations for rust-resistant cultivars. Although this could reflect cultivar-specific cup quality nutrient limitations, such simple relationships may not be reliable under variable management practices in this region of Central America. More broadly, our results demonstrate that resource-limited farmers in volcanic highlands of Guatemala currently produce high quality Arabica coffee with favorable qualities across variable management practices. Additional work is needed to constrain the relationship between yields and coffee

cup quality across a larger network of Arabica production systems with variable edaphic factors and management practices.

Acknowledgements This work was supported in part by the University of Illinois Urbana-Champaign College Agricultural, Consumer & Environmental Science Office of International Programs Global Food Security Graduate Fellowship (HAA, AJM), and CRDF Global Award no. 63777 (AJM), and no. 64622 (AJM), and the University of Illinois Urbana-Champaign Department of Crop Sciences. Logistic and community support was provided by ECA Montellano. Logistic and community support was provided by ECA Montellano.

Author's contribution AM and HA conceptualized the project; CK and AM wrote the original manuscript; CK prepared figures and tables; TB, HA and JC provided field data collection; TB, HA and DB helped with manuscript drafts. All authors reviewed the manuscript.

Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors have no competing interests to declare.

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